In [1]:
```
%load_ext autoreload
%autoreload 2
```

Asignment 02 - Random Graphs (Erdos-Reyni)

3.2. Generating Erdős-Rényi Networks

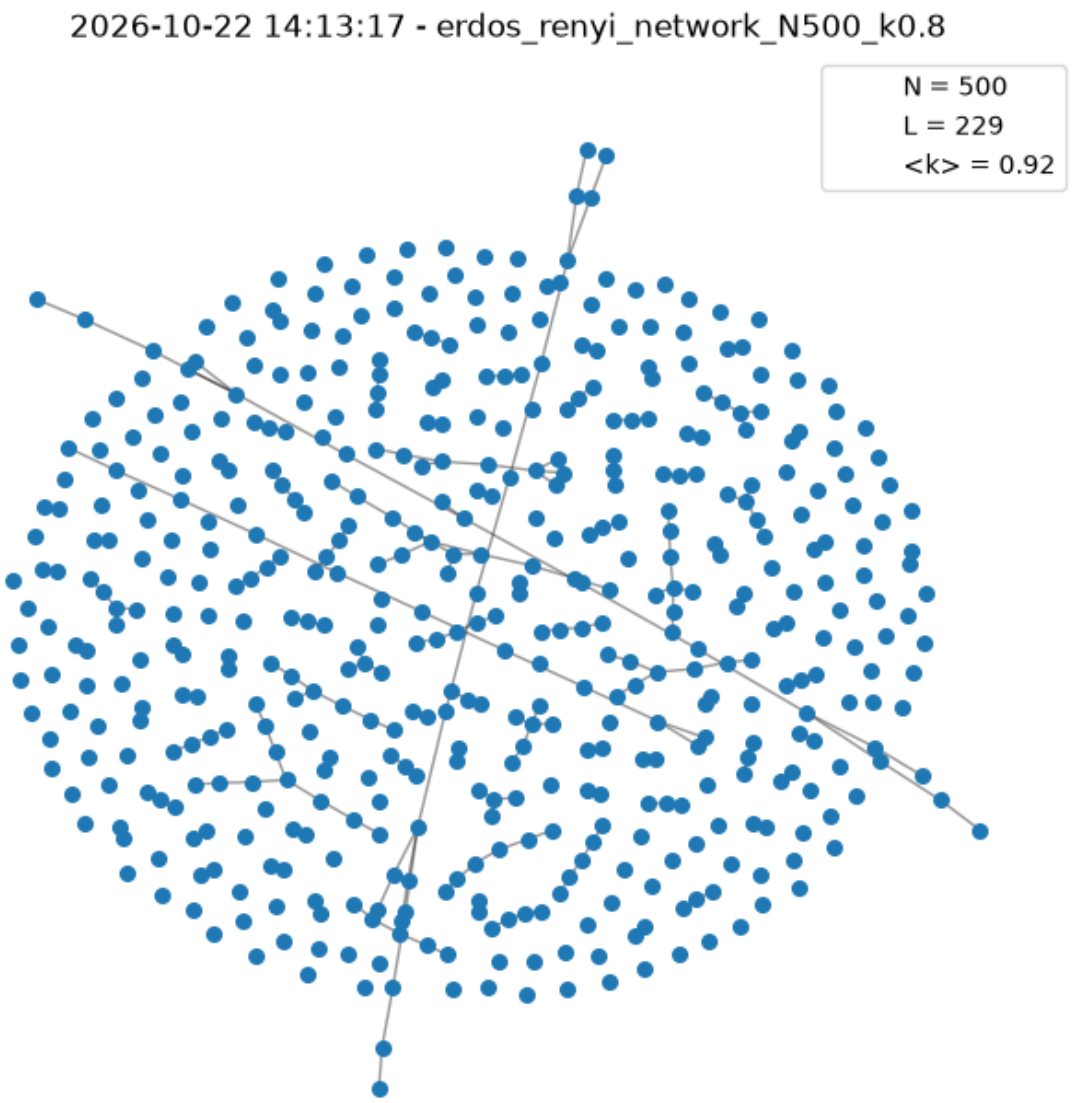
Relying on the $G(N, p)$ model, generate with a computer three networks with $N = 500$ nodes and average degree (a) $\langle k \rangle = 0.8$, (b) $\langle k \rangle = 1$ and (c) $\langle k \rangle = 8$. Visualize these networks.

In [2]:
```
%matplotlib inline
from erdos_renyi_nb import create_erdos_renyi
```

(a) $\langle k \rangle = 0.8$

In [5]:
```
G, fig = create_erdos_renyi(500, 0.8, seed=1)
```

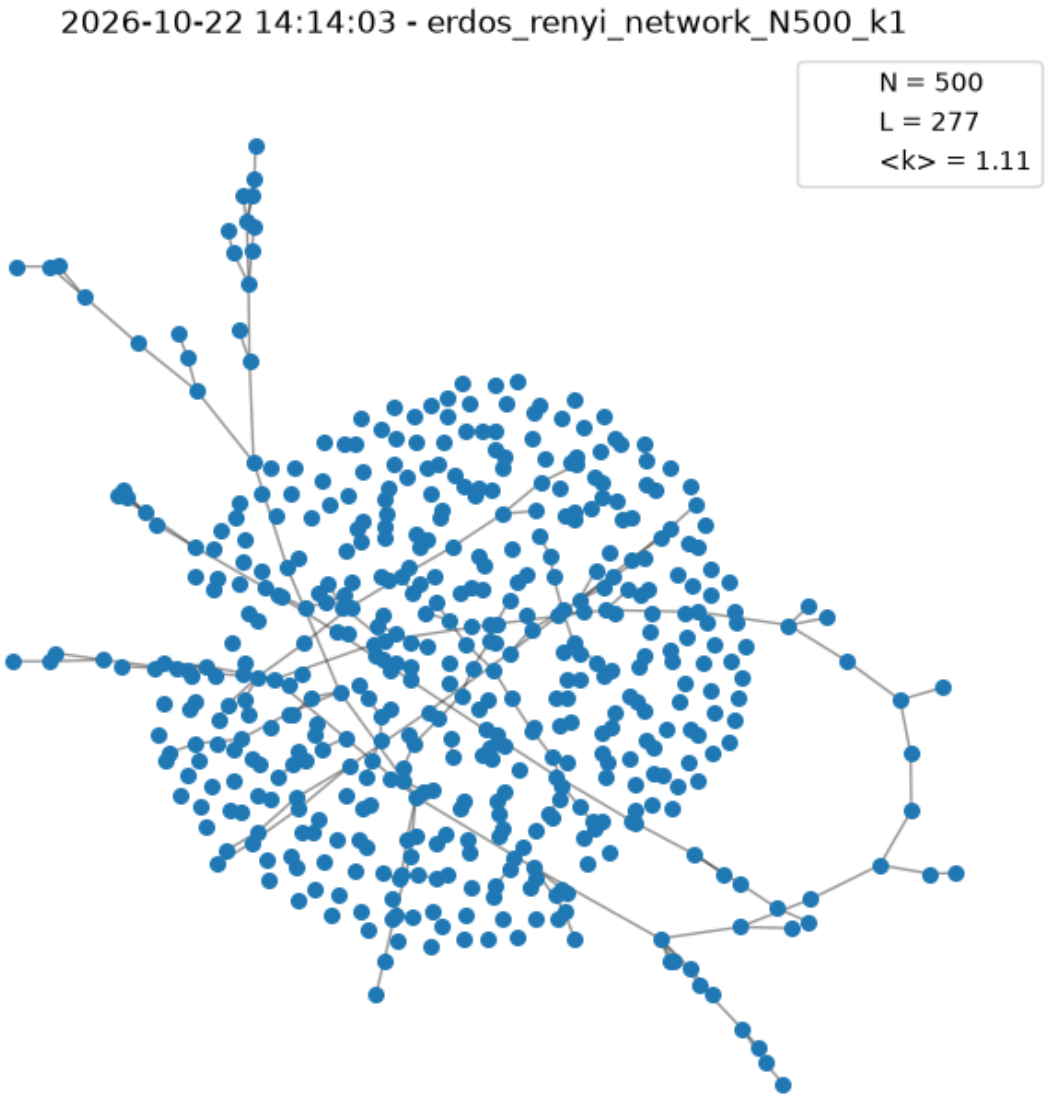
```
Saved ./20261022_141317_erdos_renyi_network_N500_k0.8.png
```



(b) $\langle k \rangle = 1$

In [6]:
```
G, fig = create_erdos_renyi(500, 1, seed=1)
```

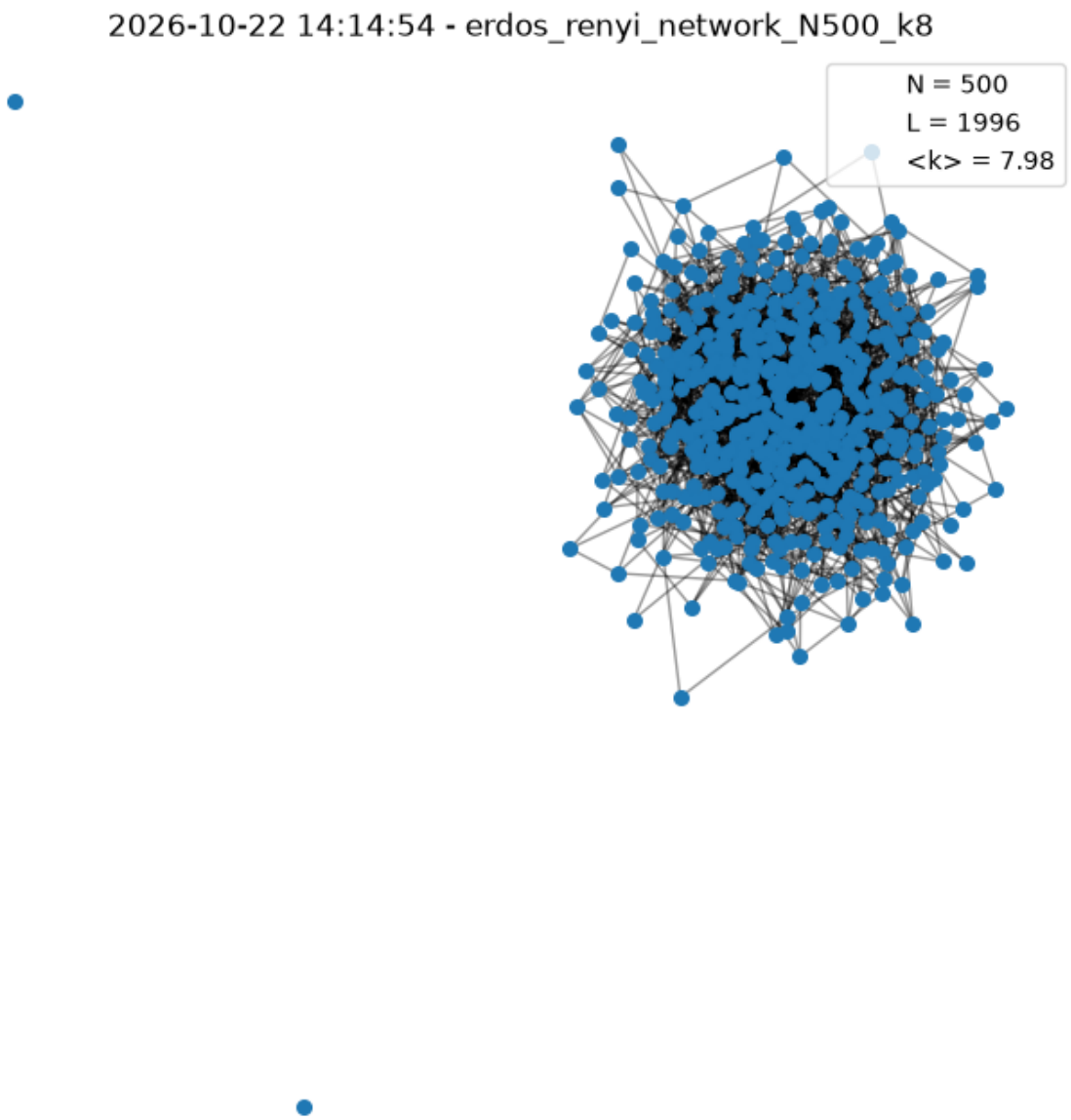
```
Saved ./20261022_141403_erdos_renyi_network_N500_k1.png
```



(c) $\langle k \rangle = 8$

In [7]:
```
G, fig = create_erdos_renyi(500, 8, seed=1)
```

```
Saved ./20261022_141454_erdos_renyi_network_N500_k8.png
```


2026-10-22 14:14:54 - erdos_renyi_network_N500_k8
N = 500
L = 1996
<k> = 7.98